MANAN RASTOGI

AI Product Engineer | Applied CV, LLM Evaluation, Full-Stack Systems

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EDUCATION

Vellore Institute of Technology (VIT), Vellore, India **2024 -- PRESENT**

B.Tech in Electronics & Communication Engineering (Biomedical Specialization)

- **Core:** Digital Signal Processing, Biomedical Imaging, Neural Networks, Embedded Systems.

IIT Madras, Online, India **2025 -- PRESENT**

BS in Data Science & Applications

- **Core:** Machine Learning Foundations, Statistical Modeling, Algorithmic Design.

TECHNICAL EXPERTISE

Applied AI & Evaluation: Python 3.12, LLM APIs (NVIDIA LLaMA 3.1), prompt engineering, RAG pipelines, rule-based validation, OpenCV, scikit-image, TrackPy, SciPy, NumPy.

Web & Full-Stack: TypeScript, React.js, Next.js, Node.js, Three.js (R3F), Framer Motion, Flask, RESTful APIs, Tailwind CSS, MongoDB, Vite.

Systems & Tooling: Git, Docker, Vercel, MongoDB Atlas, SQLite, Tesseract OCR, CI/CD, Linux Bash.

ECE & Embedded: Verilog, C++, digital logic, DSP fundamentals, embedded systems, microcontrollers.

TECHNICAL PROJECT DEPLOYMENTS

WoundTrack AI: Applied Computer Vision Pipeline **2024 -- PRESENT**

Python, OpenCV, Flask, SciPy, SQLite **[Live Demo] [Source]**

- Engineered a research-grade CV pipeline tracking **441 cells** across **800+ frames** in **<10 min** on consumer hardware, achieving **28% higher precision** than manual ImageJ workflows.
- Implemented LoG blob detection + LAP gap-closing tracker with Kalman prediction for explainable cell localization and trajectory continuity through missed frames.
- Designed custom evaluation metrics (MSD exponents, wavefront roughness, fractal-dimension analysis) to validate output against biological movement patterns.
- Built Flask + SQLite dashboard for batch processing, kinetic visualization, and automated report generation.

VITalize AI: Full-Stack Academic Platform **2025 -- PRESENT**

Next.js 15, TypeScript, MongoDB, NVIDIA LLaMA 3.1, Tesseract OCR **[Live Demo] [Source]**

- Built a unified academic platform indexing **3,000+** past exam papers with full-text search, subject-level filters, and bulk download — replacing scattered Telegram groups and Google Drive folders for VIT's student body.
- Integrated **NVIDIA LLaMA 3.1** API with Tesseract OCR fallback pipeline to extract text from scanned PDFs and generate unit-wise, mark-weighted AI solutions with KaTeX rendering.
- Engineered constraint-based **FFCS timetable solver** with clash elimination, faculty prioritization, gap minimization, and ICS calendar export.
- Deployed on **Vercel** with **MongoDB Atlas**; built a Chrome extension for VTOP data sync; open-source with community funding.

Interactive 3D Engineering Portfolio **2024 -- PRESENT**

TypeScript, React, Three.js (R3F), Framer Motion, Tailwind CSS **[Live Demo] [Source]**

- Built a portfolio with **React Three Fiber** 3D background scene and **Framer Motion** scroll-triggered section reveals.
- Achieved **WCAG accessibility** with skip navigation, ARIA labels, keyboard-dismissible intro, and reduced-motion support.
- Responsive design with lazy-loaded 3D scene, chunk-split bundling, and scroll progress indicator.

Editron — Open Source Contributor (GSSoC '26) **2026**

JavaScript, Git, Open Source Workflow

- Contributing code-quality improvements to Editron, an open-source code editor, through GirlScript Summer of Code 2026.
- Following standard open-source workflow: issue triage, fork, feature branch, and pull request.